

Quantization of rank- r matrices, with application to Discrete Cosine Transform and Image Compression

Maël Chaumette Rémi Gribonval Elisa Riccietti

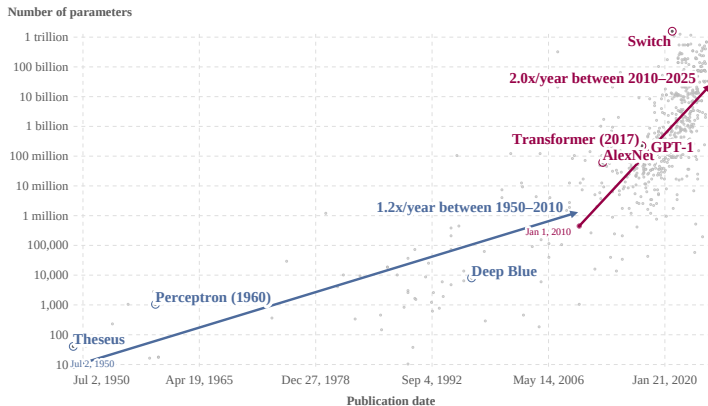
OCKHAM Team

STructured OPTimization for inverse and learning problems

01/10/2026



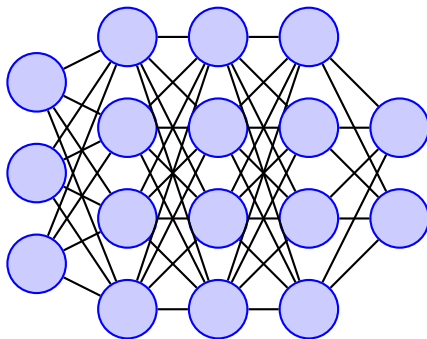
Context



Explosion in the size of deep learning models.
Source: [Samborska, 2025]

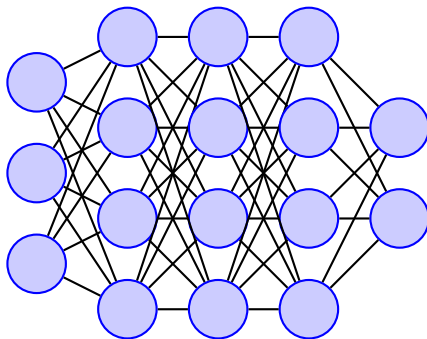
Objective

- **Main goal:** Quantize the weights of a neural network



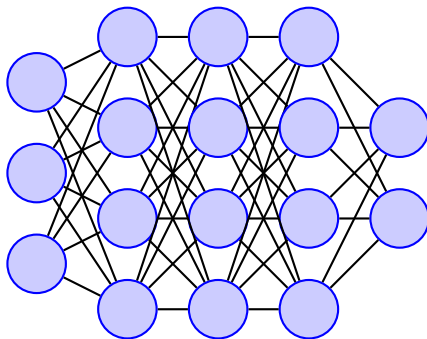
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- **Naive approach:** Map each weight to its nearest neighbor in the quantization set



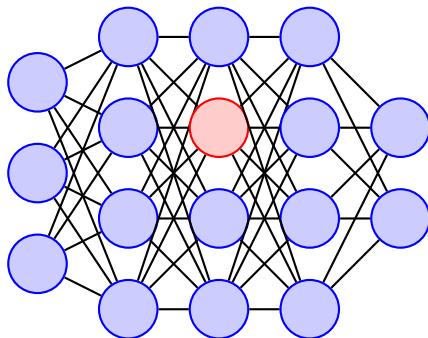
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- **Opportunity:** Take into account the *rescaling invariance* property of ReLU neural networks [Neyshabur et al., 2015]



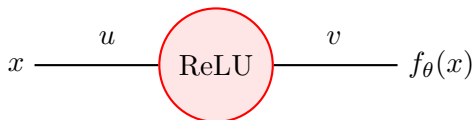
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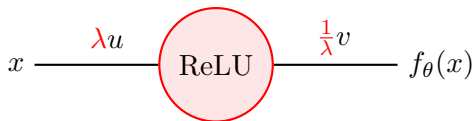
What is the rescaling invariance property?

- $\theta = (u, v) \in \mathbb{R}^{m+n}$
- $f_\theta : x \in \mathbb{R}^m \mapsto \text{ReLU}(\langle u, x \rangle)v \in \mathbb{R}^n$

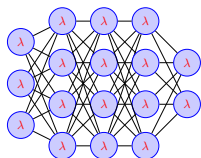


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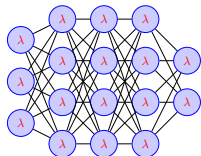


Outline



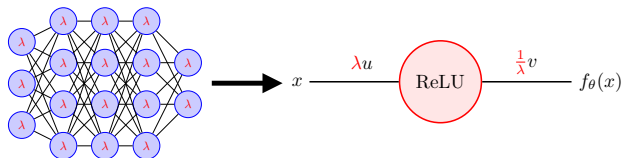
Outline

 too many λ 's



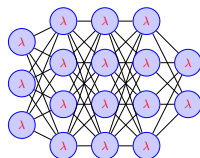
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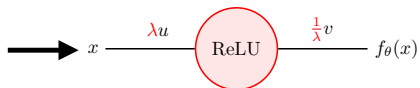


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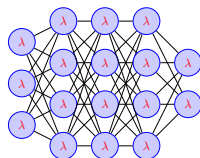


 non-linearity

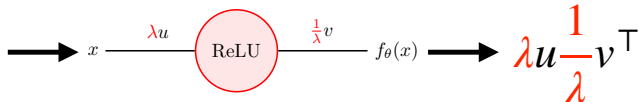


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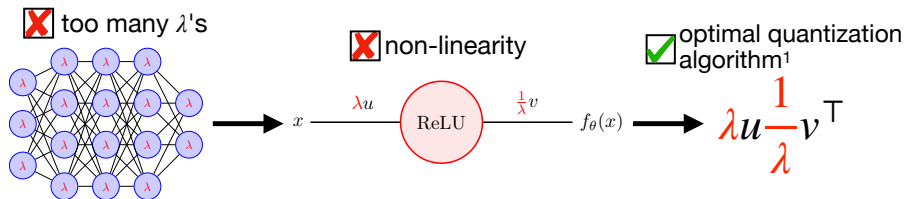
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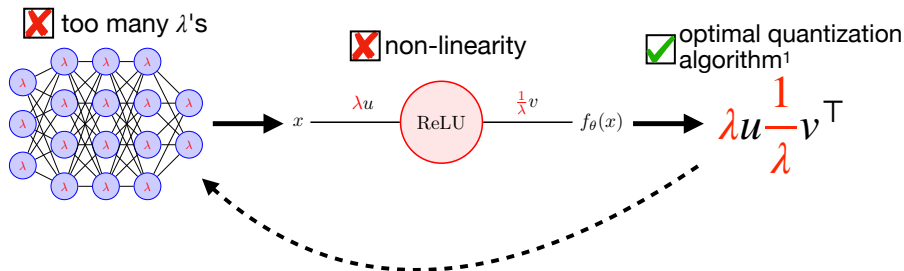


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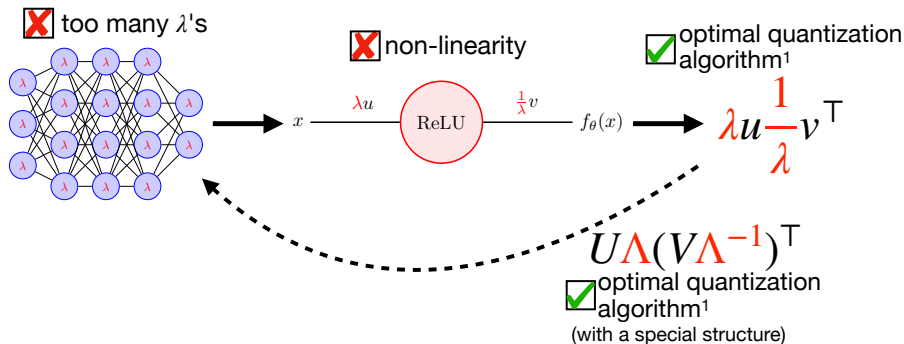
¹Proved by Gribonval et al. [2023]

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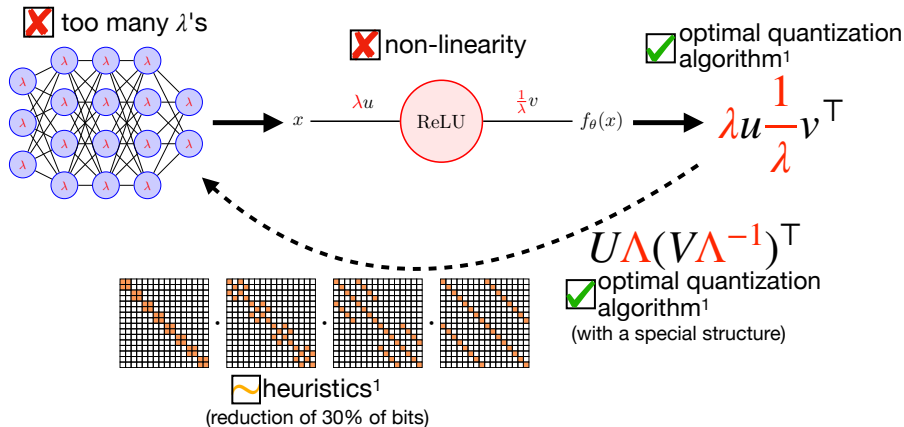
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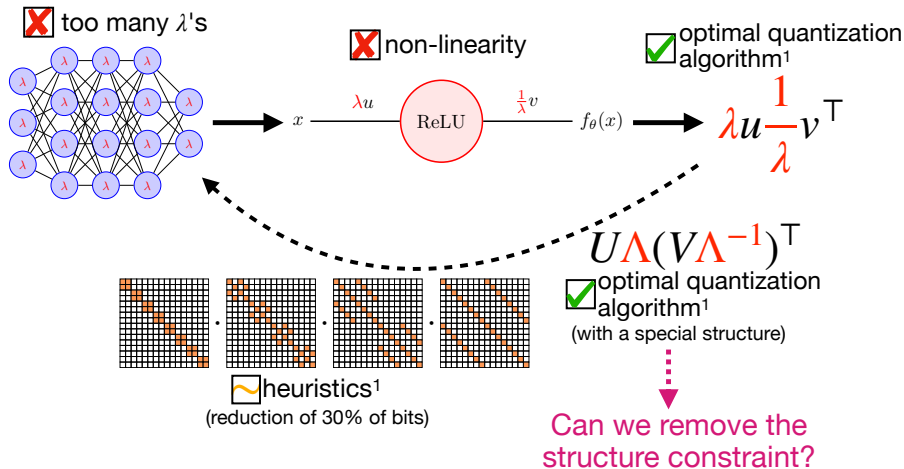
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Quantization of rank- r matrices

$$\forall \Lambda \in \text{GL}_r(\mathbb{R}), \quad (X \Lambda^\top) (Y \Lambda^{-1})^\top$$

Problem statement

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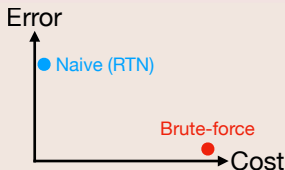
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Potential approaches



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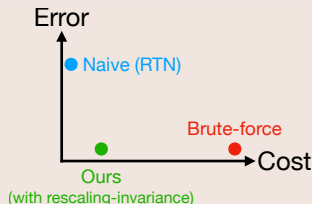
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Potential approaches



Problem resolution (for $r \geq 1$)

First lemma: reduction of the error

Given $(x, y) \in \mathbb{R}^m \times \mathbb{R}^n$, consider any $(\hat{x}, \hat{y}) \in \mathbb{F}_t^m \times \mathbb{F}_t^n$:

$$C_{x,y}(\hat{x}, \hat{y}) \geq C_{x,y}(\hat{x}', \hat{y}) \quad \forall \hat{x}' = \text{round}(\lambda x)$$

where

- λ is a simple function of $\hat{y}$ and y
- $\text{round} := \arg \min_{\hat{u}} \|\cdot - \hat{u}\|$

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 Minimizing a Mahalanobis norm is **NP-hard** [Ajtai, 1998].

Proposed heuristics for the problem

Heuristic 1: r optimal rank-one quantizations

We write

$$\|\mathbf{XY}^\top - \hat{\mathbf{X}}\hat{\mathbf{Y}}^\top\|^2 = \sum_{i=1}^r \|x_i y_i^\top - \hat{x}_i \hat{y}_i^\top\|^2 + \text{Cross Terms}$$

→ Neglect the cross terms and apply r times the rank-one quantization algorithm.

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Heuristic 2: greedy rank-one quantization

- 1 We define $R_1 := XY^\top$
- 2 For $i = 1, \dots, r$:
 - 1 $\tilde{x}_i, \tilde{y}_i \in \arg \min \|R_i - x_i y_i^\top\|^2$ via truncated SVD
 - 2 $\hat{x}_i, \hat{y}_i \in \arg \min \|\tilde{x}_i \tilde{y}_i^\top - \hat{x}_i \hat{y}_i^\top\|^2$ via rank-one quantization algorithm
 - 3 Update $R_{i+1} := R_i - \hat{x}_i \hat{y}_i^\top$

Comparison with the baseline

Metric definition:

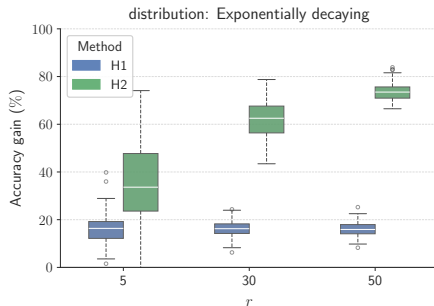
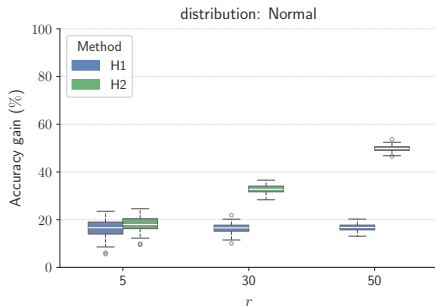
$$\rho(\hat{x}, \hat{y}) := \|\textcolor{blue}{xy}^\top - \hat{x}\hat{y}^\top\| / \|\textcolor{blue}{xy}^\top\|$$

$$\text{Accuracy Gain} := 100 \cdot \left(1 - \frac{\rho}{\rho_{\text{rtn}}}\right)$$

Experimental setup: We generate 100 pairs (X, Y) with $m = n = 64$ and $t = 4$.

- Each matrix X and Y is generated from a normal distribution.
- $XY^\top = U \text{diag}(\sigma_1, \dots, \sigma_r) V^\top$ with U, V orthogonal and $\sigma_i \sim \mathcal{E}(1)$.

Comparison with the baseline



→ Heuristic 2 is more accurate than heuristic 1 and RTN, especially when the rank increases and the singular values of XY^T are exponentially decreasing.

Application to DCT quantization

What's the DCT?

Discrete Cosine Transform (DCT)

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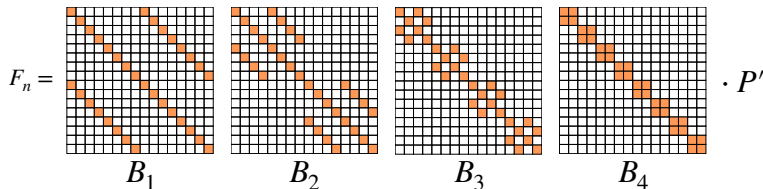
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F_n admits a butterfly factorization [Cooley and Tukey, 1965].



The diagram illustrates the butterfly factorization of the Fourier matrix F_n . It shows the equation $F_n = B_1 B_2 B_3 B_4 \cdot P'$. Each B_i is represented by an 8x8 grid where non-zero elements are highlighted in orange. B_1 shows a single diagonal of orange squares. B_2 shows two parallel diagonals. B_3 shows four parallel diagonals. B_4 shows eight parallel diagonals, representing a sparse matrix. The permutation matrix P' is indicated by a dot and the label P' to the right of the final grid.

Does the DCT admit a butterfly factorization?

Therefore, we get

$$C_n = \text{Re}(B_1 \cdots B_L) P' P$$

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→ Fortunately, we prove that if a matrix A admits a butterfly factorization, then $\operatorname{Re}(A)$ also admits a butterfly factorization but **different factors**:

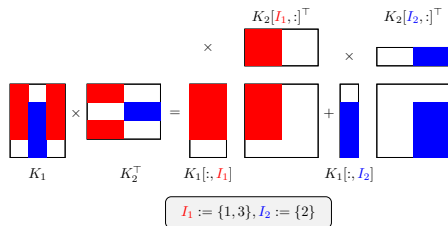
$$\operatorname{Re}(A) = K_1 \cdots K_L$$

Quantization of Butterfly factors

If $L = 2$: we have

$$K_1 K_2^\top = \sum_{I \in \mathcal{P}} K_1[:, I] K_2[:, I]^\top$$

where the $K_1[:, I] K_2[:, I]^\top$ have **disjoint support**.

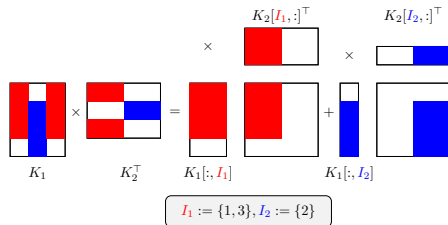


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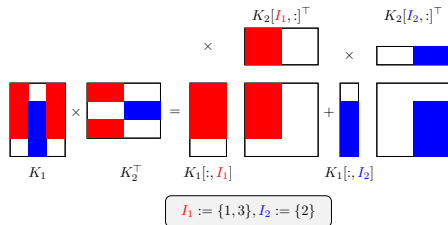
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If $L > 2$, we can write $K_1 \cdots K_L = (K_1 K_2)(K_3 K_4) \cdots (K_{L-1} K_L)$ and apply the same reasoning as $L = 2$.

Experimental results

Metric definition:

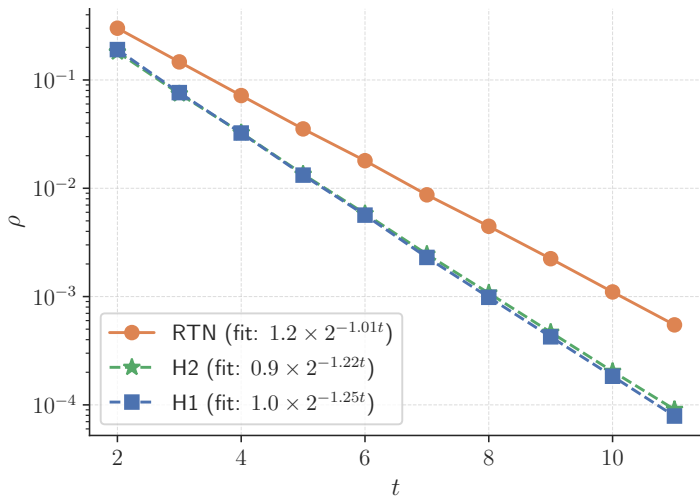
$$\rho := \|K_1 \cdots K_L - \hat{K}_1 \cdots \hat{K}_L\| / \|K_1 \cdots K_L\|$$

$$\text{Accuracy Gain} := 100 \cdot \left(1 - \frac{\rho}{\rho_{\text{rtn}}}\right)$$

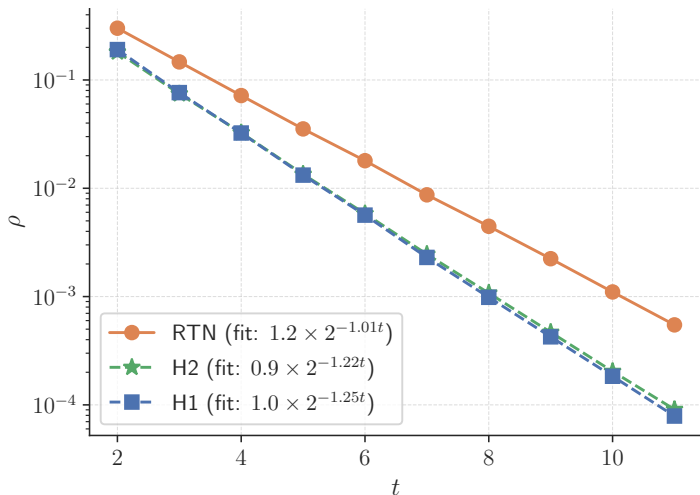
Experimental setup:

We generate 100 random butterfly matrices of size $n = 1024$ and try to quantize the factors appearing when taking the real part.

Experimental results

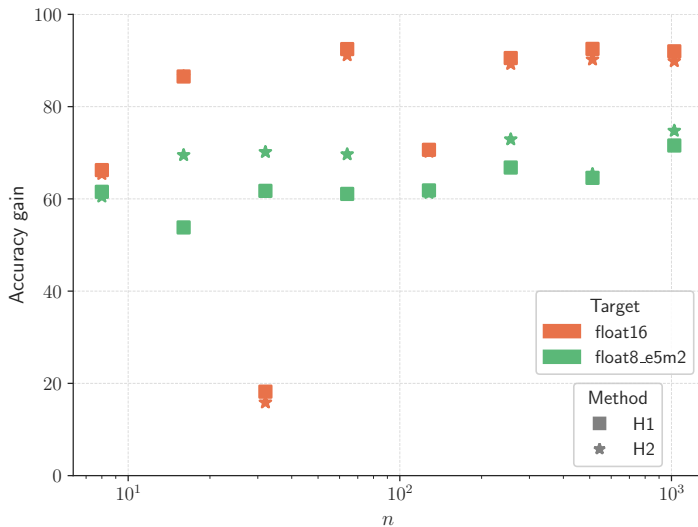


Experimental results



→ Reduction of 20% of bits compared to RTN.

On the DCT



Wrap-up:

- Accurate algorithms to quantize a product of matrices of any rank
- Compared to RTN, the number of bits is reduced by 20% for a given precision on butterfly matrices
- Accurate compression/quantization algorithm on images

What's next?

- Extend this work to quantize ReLU networks

Thanks for your attention

References I

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<https://ourworldindata.org/scaling-up-ai>.
- [2] B. Neyshabur, R. Tomioka, and N. Srebro. Norm-based capacity control in neural networks. In *Conference on learning theory*, pages 1376–1401. PMLR, 2015.
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- [4] M. Ajtai. The shortest vector problem in l_2 is np-hard for randomized reductions (extended abstract). In *Proceedings of the Thirtieth Annual ACM Symposium on Theory of Computing*, page 10–19, 1998.
- [5] J. Makhoul. A fast cosine transform in one and two dimensions. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 28:27–34, 1980.

- [6] J. W. Cooley and J. W. Tukey. An algorithm for the machine calculation of complex fourier series. *Mathematics of computation*, 19(90):297–301, 1965.
- [7] P. Ashtari, P. Behmandpoor, F. Nateghi Haredasht, J. H. Chen, P. Patrinos, and S. Van Huffel. Quantization-aware matrix factorization for low bit rate image compression. *Information Sciences*, 722:122646, 2026.

Expressions for the rescaling-invariant part

Expression of f :

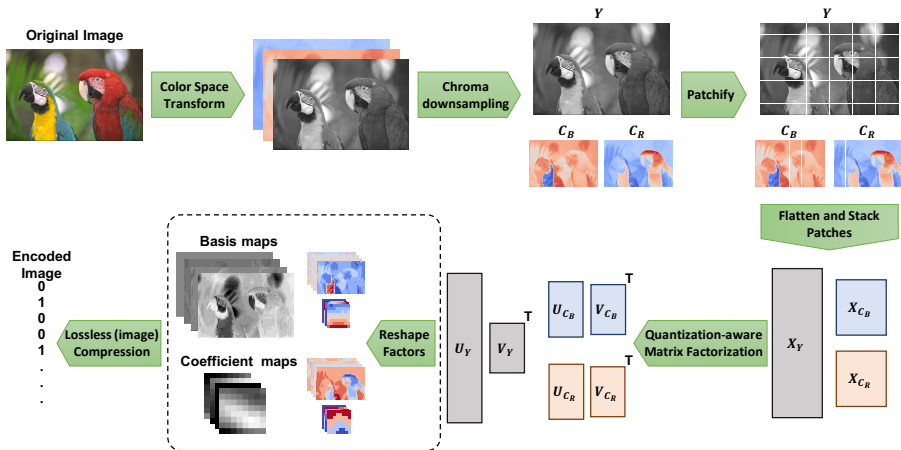
$$f : \lambda \in \mathbb{R} \mapsto \max_{\hat{x} \in \text{round}(\lambda x)} \|xy^\top - \hat{x} \text{round}(\mu(\hat{x})y)^\top\|$$

where $\mu(\hat{x}) := \frac{\langle \hat{x}, x \rangle}{\|\hat{x}\|^2}$ if $\hat{x} \neq 0$ and 0 otherwise.

Expression of Λ :

$$\Lambda := \hat{Y}^\dagger Y$$

On images (Compression setup)



We apply the same pre-processing process as Ashtari et al. [2026]

On images (Experimental results)

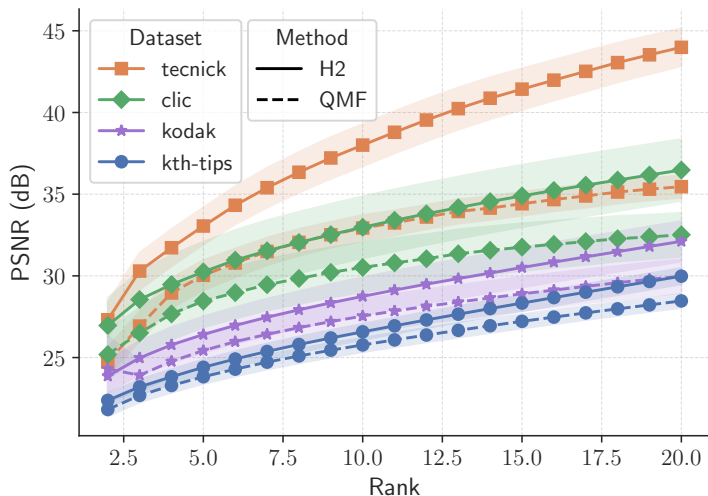
Metric definition:

We use the Peak Signal-to-Noise Ratio (PSNR) to measure the quality of the reconstructed image.

Experimental setup:

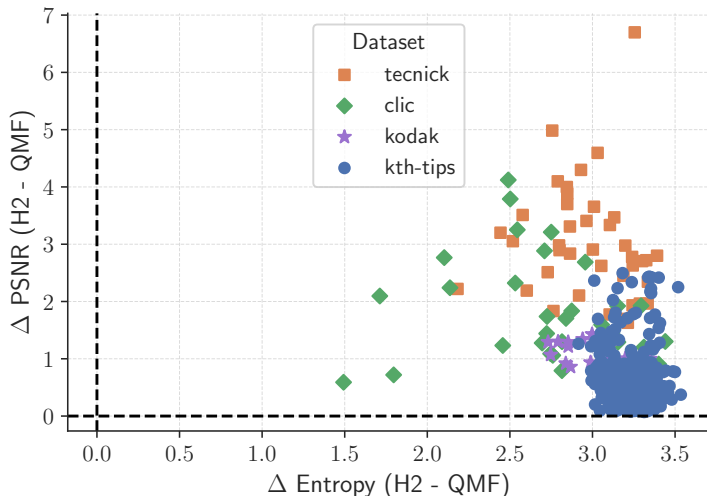
We use four different datasets: Kodak, Clic validation, KTH-TIPS and tecnick. We quantize each factor in `float8_e4m3`

On images (Experimental results)



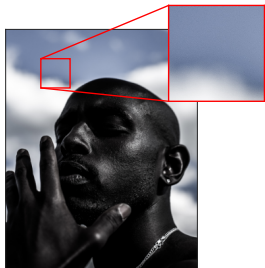
Our method outperforms QMF on every dataset.

On images (Experimental results)

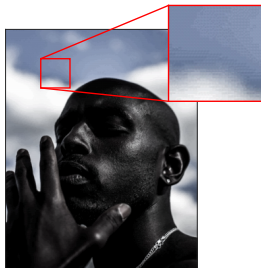


However, our method has a higher entropy than QMF.

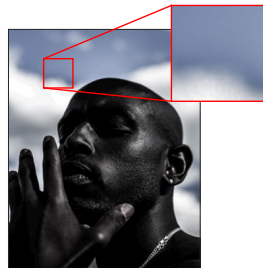
On images (Experimental results)



Ground Truth



PSNR: 25.99 dB, Entropy: 2.35
QMF



PSNR: 26.71 dB, Entropy: 4.15
H2

The background reconstructed by QMF contains visible artifacts.